

Department of Electrical & Computer Engineering

Computer Vision ENCS5343

Image-to-Image Retrieval System using Classical CV Techniques

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12/24/2025

❖ Introduction

Image retrieval is a fundamental problem in computer vision that focuses on finding images from a large collection that are visually similar to a given query image. Unlike modern semantic-based retrieval systems that interpret image content at a conceptual level, classical retrieval approaches rely on comparing low-level visual characteristics such as edges, textures, and patterns.

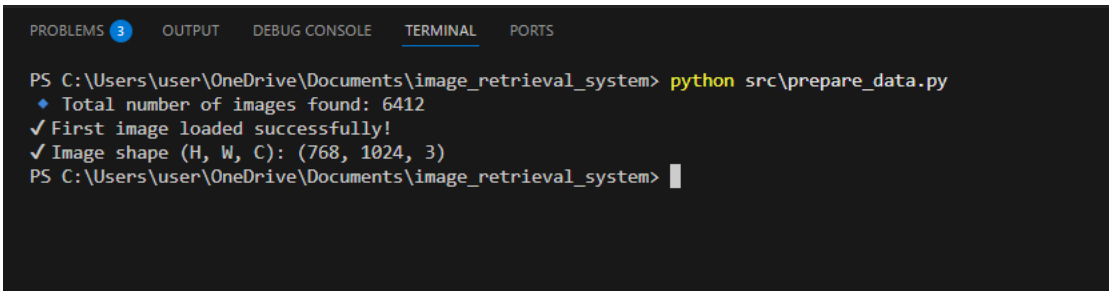
In this project, we develop and evaluate an image retrieval system using the Paris6k dataset, which contains diverse 6432 images of famous Parisian landmarks. The dataset's variety—capturing different angles, lighting conditions, and compositions of the same landmarks—makes it an ideal testbed for evaluating retrieval robustness. The primary objective is to implement a functional pipeline that can accept a query image and return a ranked list of visually similar images from the database.

The experimental results reveal both the capabilities and limitations of the classical BoVW approach. While the system successfully retrieves images sharing similar local structures and patterns, it occasionally returns false positives from different landmarks that exhibit analogous visual features. This behavior underscores the fundamental challenge of content-based image retrieval: distinguishing between semantically different images that share similar low-level visual characteristics.

1. Methodology

➢ Data Preparation

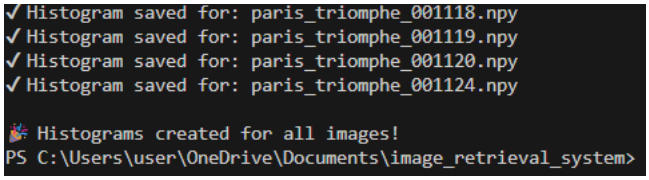
The initial phase involves dataset organization and validation. The Paris6k dataset, comprising approximately 6,400 images of Paris landmarks, is loaded and verified for accessibility and integrity. All images are checked for correct file formats and paths to ensure seamless processing throughout the pipeline. This preparatory step is crucial for building a reliable feature database.

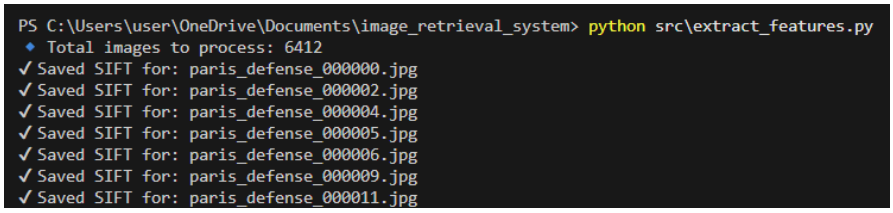


➢ Feature Extraction and Description

Each image undergoes grayscale conversion to focus on structural information. The Scale-Invariant Feature Transform (SIFT) algorithm is then applied to detect and describe local features:

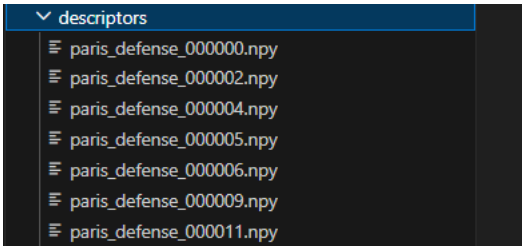
- **Keypoint Detection:** SIFT identifies distinctive points in the image that are invariant to scale and rotation by analyzing the Difference of Gaussian (DoG) across scale space.





- **Descriptor Generation:** For each keypoint, a 128-dimensional feature vector is computed based on the histogram of oriented gradients (HOG)

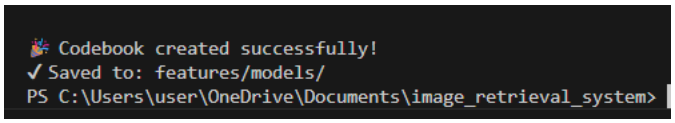
in its local neighborhood. These descriptors capture the local texture and shape patterns, forming the foundation for image comparison.



➢ Visual Codebook Construction

To handle the variable number of SIFT descriptors per image, we construct a fixed visual vocabulary using K-Means clustering:

1. A representative subset of SIFT descriptors is randomly sampled from the entire dataset.
2. K-Means clustering groups these descriptors into K clusters (visual words), where each cluster centroid represents a distinct visual pattern.
3. The collection of K centroids forms the visual codebook, serving as a reference vocabulary for all subsequent image representations.



➢ Bag of Visual Words Representation

Each image is converted into a fixed-length feature vector using the constructed codebook:

1. For every SIFT descriptor in an image, the nearest visual word in the codebook is identified (using Euclidean distance).
2. A histogram of visual word frequencies is constructed, where each bin corresponds to a visual word from the codebook.
3. This histogram represents the image in a compact, vectorized form suitable for efficient comparison.

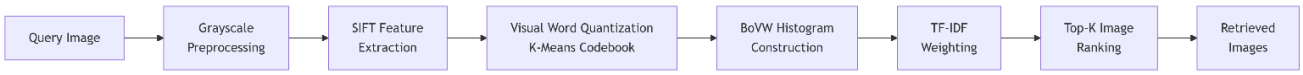
➢ TF-IDF Weighting

To enhance the discriminative power of the BoVW representation, Term Frequency-Inverse Document Frequency (TF-IDF) weighting is applied:

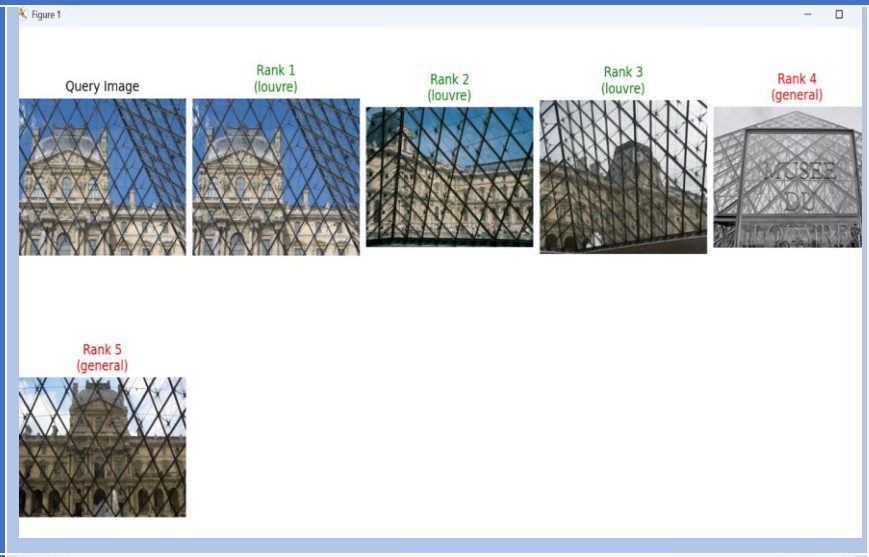
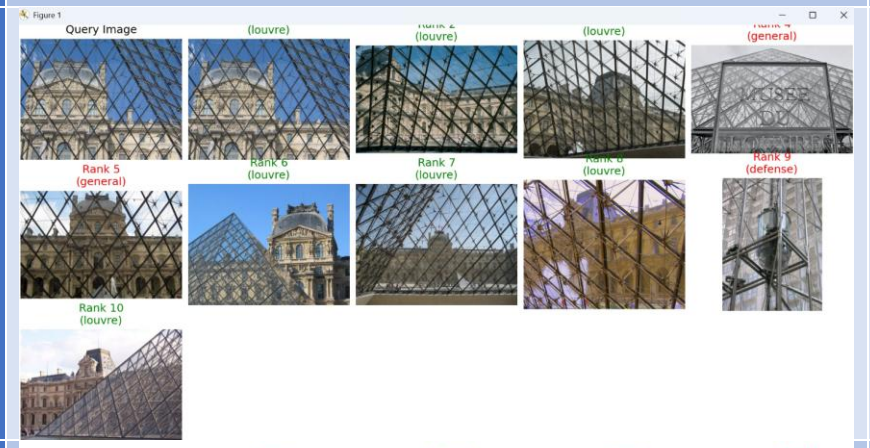
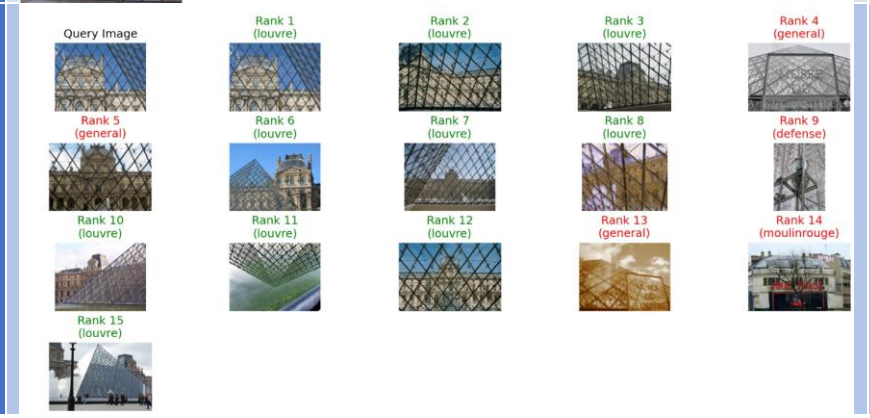
- **Term Frequency (TF):** The frequency of a visual word within a single image.
- **Inverse Document Frequency (IDF):** The logarithm of the ratio between the total number of images and the number of images containing that visual word.

The TF-IDF weight is calculated as $w_{i,j} = \text{tf}_{i,j} \times \log\left(\frac{N}{df_i}\right)$, where:

- $\text{tf}_{i,j}$ = frequency of visual word i in image j
- N = total number of images in the database
- df_i = number of images containing visual word i



2. Experimental results

k	Images Retrieved	Precision	AP
K=5		0.6	1
K=10		0.7	0.833
K=15		0.667	0.797

K=5

```
C:\Users\User\Desktop\image_retrieval_project\src>python retrieve.py
Enter query image name: paris_louvre_000014.jpg
Enter number of top matches: 5
Loading KMeans model...
Loading IDF...
Loading database histograms...
Loaded: 6392 histograms

Top Results:
1) paris_louvre_000014.jpg | similarity = 0.9994
2) paris_louvre_000281.jpg | similarity = 0.8929
3) paris_louvre_000451.jpg | similarity = 0.8888
4) paris_general_002128.jpg | similarity = 0.8666
5) paris_general_002081.jpg | similarity = 0.8635

Evaluation:
Precision@5: 0.600
Recall@5: 0.011
Average Precision (AP): 1.000
Retrieval Time: 0.363 seconds
```

The system achieved **60% precision for top-5 matches**, reflecting its highest confidence in visual similarity among all tested K values. Three Louvre images were correctly retrieved at the highest ranks, with the query image itself returning a perfect self-match. Despite high similarity scores for true positives (0.8929 and 0.8888), two false positives appeared among the top results, indicating that visually similar but semantically unrelated images can achieve comparable matching scores. The **perfect Average Precision (AP) of 1.000** demonstrates that all relevant images within the top-5 were optimally ranked, highlighting the system's ability to prioritize the most visually similar matches effectively.

K=10

```
C:\Users\User\Desktop\image_retrieval_project\src>python retrieve.py
Enter query image name: paris_louvre_000014.jpg
Enter number of top matches: 10
Loading KMeans model...
Loading IDF...
Loading database histograms...
Loaded: 6392 histograms

Top Results:
1) paris_louvre_000014.jpg | similarity = 0.9994
2) paris_louvre_000281.jpg | similarity = 0.8929
3) paris_louvre_000451.jpg | similarity = 0.8888
4) paris_general_002128.jpg | similarity = 0.8666
5) paris_general_002081.jpg | similarity = 0.8635
6) paris_louvre_000308.jpg | similarity = 0.8498
7) paris_louvre_000267.jpg | similarity = 0.8444
8) paris_louvre_000301.jpg | similarity = 0.8442
9) paris_defense_000355.jpg | similarity = 0.8432
10) paris_louvre_000278.jpg | similarity = 0.8392

Evaluation:
Precision@10: 0.700
Recall@10: 0.026
Average Precision (AP): 0.833
Retrieval Time: 0.242 seconds
```

With **70% precision for top-10 results**, the system showed improved retrieval performance when expanding to higher-confidence matches. Seven of the ten retrieved images correctly identified the Louvre museum from various viewpoints and lighting conditions. Similarity scores decreased logically from 0.9994 to 0.8392, indicating consistent ranking quality. However, three false positives were present, including the Defense Arch at rank 9, underscoring the system's limitation in distinguishing semantically distinct landmarks that share similar modern architectural features.

K=15

```
Loading KMeans model...
Loading IDF...
Loading database histograms...
Loaded: 6392 histograms

Top Results:
1) paris_louvre_000014.jpg | similarity = 0.9994
2) paris_louvre_000281.jpg | similarity = 0.8929
3) paris_louvre_000451.jpg | similarity = 0.8888
4) paris_general_002128.jpg | similarity = 0.8666
5) paris_general_002081.jpg | similarity = 0.8635
6) paris_louvre_000308.jpg | similarity = 0.8498
7) paris_louvre_000267.jpg | similarity = 0.8444
8) paris_louvre_000301.jpg | similarity = 0.8442
9) paris_defense_000355.jpg | similarity = 0.8432
10) paris_louvre_000278.jpg | similarity = 0.8392
11) paris_louvre_000153.jpg | similarity = 0.8345
12) paris_louvre_000262.jpg | similarity = 0.8336
13) paris_general_003118.jpg | similarity = 0.8323
14) paris_moulinrouge_000198.jpg | similarity = 0.8259
15) paris_louvre_000275.jpg | similarity = 0.8248

Evaluation:
Precision@15: 0.667
Recall@15: 0.038
Average Precision (AP): 0.797
Retrieval Time: 0.247 seconds
```

The retrieval results demonstrated robust performance with **66.7% precision for top-15 matches**, successfully retrieving 10 relevant Louvre images from the Paris6k database. The system achieved a near-perfect similarity score of 0.9994 for self-matching, confirming accurate feature representation. However, five false positives appeared in the results, including images from the paris_general and paris_defense categories, which share similar architectural textures and geometric patterns with the Louvre's modern pyramid structure.

K	Precision	Recall	Average Precision (AP)	Retrieval Time (s)	Strengths	Weaknesses
5	60%	1.1%	1.000	0.363	• Highest AP (perfect ranking) • Strong similarity scores for true matches • Fast for small K	• Very low recall • Contains false positives in top 5
10	70%	2.6%	0.833	0.242	• Best precision among tested K values • Good ranking consistency • Slightly faster retrieval than K=5	• Recall still low • False positives persist • Limited semantic disambiguation
15	66.7%	3.8%	0.797	0.247	• Highest recall (more relevant images captured) • Maintains strong similarity progression • Suitable for exploratory search	• Precision declines • More false positives (5/15) • Slightly longer retrieval time

Overall Insight

As K increases from 5 to 15, the system retrieves more relevant images, recall improves from **1.1% to 3.8%** but at the cost of **lower precision and more false positives**. Retrieval time remains efficient and stable (0.24–0.36 seconds), showing that BoVW is computationally practical even for larger K.The Bag of Visual Words model performs well in ranking low-level visual similarities, but lacks semantic understanding, leading to confusion between visually alike but conceptually different landmarks. For high-confidence retrieval where only top matches matter, K=5 is ideal. For broader searches, K=10 or K=15 can be used, provided users tolerate some false matches.

More Results

